

PRACTICEL NO. 2

*//2) design an application by using an array -->
//HANDS TOSSING GAME IN JAVA*

```
package practicles;
import java.util.Random;
import java.util.Scanner;

public class design_app_by_array_2 {
    public static void main(String[] args) {
        String[] names;      //declaring an String array
        Scanner sc = new Scanner(System.in);
        Scanner sc1 = new Scanner(System.in);
        int score = 0;
        int tmp_score = 0;
        System.out.println("enter the number of players. You cannot add less than
3 players and more than 10 players");

        int no_of_players;
        do{
            no_of_players = sc.nextInt();
            if(no_of_players<10 && no_of_players>3){
                System.out.println("you will add "+no_of_players+" of players");
            }
            else if(no_of_players>10){
                System.out.println("you can add only upto 10 players");
            }
            else if(no_of_players<3 ){
                System.out.println("you can not add less than 3 players");
            }
        }while (no_of_players>10 || no_of_players<3);

        names = new String[no_of_players]; //initializing an array by the size
of "no_of_players"
        System.out.println("enter the name of players : ");
        for(int i=0;i<names.length;i++){
            System.out.println("enter the "+i+"st name");
            names[i] = sc1.nextLine();
        }

        Random r = new Random();
        int random_index;
        int n = 4;
        char toss;
        while (n>0){
            System.out.println("press n for tossing hands : ");
            toss = sc.next().charAt(0);
            if(toss == 'n'){
                random_index = r.nextInt(0,names.length);
                System.out.println("it is the turn of : "+names[random_index] +
" at index number "+random_index);
                n=n-1;
            }
            else if(toss == 'x'){
                System.out.println("finished tossing");
                break;
            }
        }
    }
}
```

```

        else {
            System.out.println("it is invalid button");
        }
    }
}

```

o/p :

enter the number of players. You cannot add less than 3 players and more than 10 players

3

enter the name of players :

enter the 0st name

john

enter the 1st name

peter

enter the 2st name

kratos

press n for tossing hands :

n

it is the turn of : peter at index number 1

press n for tossing hands :

n

it is the turn of : kratos at index number 2

press n for tossing hands :

n

it is the turn of : kratos at index number 2

press n for tossing hands :

n

it is the turn of : john at index number 0

PRACTICEL NO. 3

```
//3) implementation of package interface and abstract class
package practicles;

import java.util.Scanner;

//creating our interface
interface Student
{
    public void calculate_marks();
    public void enterStudentInfo();
    public void showStudentInfo();
    public void editStudentName(String name);
}

class IIMS implements Student{
    Scanner sc = new Scanner(System.in);
    static int initPrn = 0;
    String name, PRN, phoneNo;
    public IIMS(){
        PRN = "MC20";
    }
    public void calculate_marks(){
        int s1, s2, s3;
        System.out.println("enter the marks of subject 1:");
        s1 = sc.nextInt();
        System.out.println("enter the marks of subject 2:");
        s2 = sc.nextInt();
        System.out.println("enter the marks of subject 3:");
        s3 = sc.nextInt();

        int total = s1 + s2 + s3;
        int avg = total/3;
        System.out.println("average score is : "+avg);
    }

    public void enterStudentInfo(){
        System.out.println("enter the name of student : ");
        name = sc.nextLine();
        System.out.println("enter the phone no. of student :");
        phoneNo = sc.nextLine();

        System.out.println("your PRN no is : "+updatePRN());
    }

    public void showStudentInfo() {

    }

    public void editStudentName(String name){

    }

    public String updatePRN(){
        initPrn = initPrn + 1;
    }
}
```

```

        String tmpPRN = PRN;
        tmpPRN = tmpPRN + Integer.toString(initPrn);
        return tmpPRN;
    }
}

public class practicle_3 {
    public static void main(String[] args) {
        char select;
        int no_of_st;
        IIMS i = new IIMS();
        System.out.println("press a for entering the marks of the student");
        System.out.println("press b for student admission");
        Scanner sc = new Scanner(System.in);
        select = sc.next().charAt(0);
        switch (select){
            case 'a':
                i.calculate_marks();
            case 'b':
                System.out.println("enter the number of students for admission
:");

                no_of_st = sc.nextInt();
                while (no_of_st!=0){
                    i.enterStudentInfo();
                    no_of_st--;
                }
            default:
                System.out.println("press the valid button");
        }
    }
}

```

o/p:

press a for entering the marks of the student

press b for student admission

b

enter the number of students for admission :

2

enter the name of student :

Chaitanya

enter the phone no. of student :

90751393

your PRN no is : MC201

enter the name of student :

Sagar

enter the phone no. of student :

899874

your PRN no is : MC202

PRACTICEL NO. 4

//4) design an application using String, StringBuilder and StringTokenizer

```
package practicles;

import javax.print.attribute.EnumSyntax;
import java.util.Scanner;
import java.util.StringTokenizer;

class MyStringBuilder{
    Scanner sc = new Scanner(System.in);
    StringBuilder sb;
    public MyStringBuilder(){
        sb = new StringBuilder();
    }

    public String takeEntry(){
        System.out.println("enter the string for String Builder object");
        return sb.append(sc.nextLine()).toString();
    }

    public String showEntry(){
        return takeEntry();
    }
}

class MyStringTokenizer{
    String str;

    public MyStringTokenizer(String str){
        this.str = str;
        StringTokenizer st = new StringTokenizer(str);
        while (st.hasMoreTokens()){
            System.out.println(st.nextToken());
        }
    }
}

public class practice_4 {
    public static void main(String[] args) {
        MyStringBuilder sb = new MyStringBuilder(); //for StringBuilder Class
        sb.takeEntry();
        System.out.println("entered string in String builder object is :
"+sb.showEntry());

        MyStringTokenizer st = new MyStringTokenizer(sb.showEntry());
    }
}
```

o/p:

enter the string for String Builder object

My name is John

enter the string for String Builder object

This is next String

entered string in String builder object is : My name is JohnThis is next String

enter the string for String Builder object

raman

My

name

is

JohnThis

is

next

Stringraman

PRACTICEL NO. 5

//5) test any 5 of standard exceptions and user defined custom exceptions in java package practicles;

```
import java.io.*;

class StdExceptions{
    String name;
    int marks;
    String str;
    int ar[];
    StdExceptions(int marks, String str, int ar[]){
        this.marks = marks;
        this.str = str;
        this.ar = ar;
    }

    public void showNullPointerException(){
        try{
            if(name.equals("ram")){
                System.out.println("it is 3 letter name");
            }
            else if(name.equals("john")) {
                {
                    System.out.println("it is 4 letters name");
                }
            }
        }
        catch (NullPointerException npe){
            System.out.println(npe.getMessage());
        }
    }

    public void showArithmeticException(){
        try{
            int show = marks/0;
            System.out.println("result is : "+show);
        }
        catch (ArithmeticException arithmeticException){
            System.out.println("error : "+arithmeticException.getMessage());
        }
    }

    public void showStringIndexOUtOfBound(){
        System.out.println("select the 5 th character to print from the string
str");
        try{
            str.charAt(5);
        }catch (StringIndexOutOfBoundsException exception){
            System.out.println("error : "+exception.getMessage());
        }
    }

    public void showArrayIndexOutOfBoundException(){
        try{
            System.out.println(ar[100]);
        }
    }
}
```

```

        }catch (ArrayIndexOutOfBoundsException exception){
            System.out.println(exception.getMessage());
        }
    }

    public void showFileNotFoundException(){
        try {
            FileReader fileReader = new FileReader("C:\\text.txt");
            BufferedReader bufferedReader = new BufferedReader(fileReader);
            String read = null;
            while ((read=bufferedReader.readLine())!=null){
                System.out.println(read);
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}

class UserDefExcetion1 extends Exception{
    public UserDefExcetion1(String str){
        System.out.println("weight limit crossed");
    }
}

class UserDef1{
    int wt;
    public UserDef1(int wt) throws UserDefExcetion1 {
        this.wt = wt;
        if(wt>100){
            throw new UserDefExcetion1("weight limit crossed");
        }
        else {
            System.out.println("your weith is "+wt);
        }
    }
}

public class practicle_5 {
    public static void main(String[] args) throws UserDefExcetion1 {
        int ar[] = {89, 832, 24};

        //showing 5 standard exceptions
        StdExceptions stdExceptions = new StdExceptions(90, "John", ar);
        stdExceptions.showNullPointerException(); //NullPointerException
        stdExceptions.showArithmeticException(); //ArithmetciException
        stdExceptions.showStringIndexOUtOfBound(); //StringIndexOutOfBound
        stdExceptions.showArrayIndexOutOfBoundException();
        //ArrayIndexOUtOfBoundException
        stdExceptions.showFileNotFoundException();
        //FileNotFoundException

        //showing user defined exceptions
        UserDef1 userDef1 = new UserDef1(101); } }

```

```
o/p:
Cannot invoke "String.equals(Object)" because "this.name" is null
error : / by zero
select the 5 th character to print from the string str
error : String index out of range: 5
Index 100 out of bounds for length 3
java.io.FileNotFoundException: C:\text.txt (The system cannot find the file
specified)
    at java.base/java.io.FileInputStream.open0(Native Method)
    at java.base/java.io.FileInputStream.open(FileInputStream.java:216)
    at java.base/java.io.FileInputStream.<init>(FileInputStream.java:157)
    at java.base/java.io.FileInputStream.<init>(FileInputStream.java:111)
    at java.base/java.io.FileReader.<init>(FileReader.java:60)
    at
practicles.StdExceptions.showFileNotFoundException(practicle_5.java:62)
    at practicles.practicle_5.main(practicle_5.java:103)
weight limit crossed
Exception in thread "main" practicles.UserDefExcetion1
    at practicles.UserDef1.<init>(practicle_5.java:86)
    at practicles.practicle_5.main(practicle_5.java:106)
```

PRACTICLE NO. 6

```
//6) thread creation and design application by using extending Thread
class/implementing the Runnable interface
// application of multithreading in java
package practicles;

class printFiveMultiples extends Thread{

    public void run(){
        for(int i=0;i<5;i++){
            try{
                Thread.sleep(1000);
                System.out.println(5 * i);
            }catch (Exception e){
                e.printStackTrace();
            }
        }
    }
}

class sevenMultiples extends Thread{

    public void run(){
        for(int i=0;i<5;i++){
            try {
                Thread.sleep(1000);
                System.out.println(7 * i);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}

public class practicle_6 {
    public static void main(String[] args) {
        printFiveMultiples printFiveMultiples = new printFiveMultiples();
        sevenMultiples sevenMultiples = new sevenMultiples();
        for (int i=0;i<2;i++){
            Thread t1 = new Thread(printFiveMultiples);
            t1.start();

            Thread t2 = new Thread(sevenMultiples);
            t2.start();
        }
    }
}
```

o/p:

0

0

0

0

5

7

5

7

10

14

10

14

PRACTICEL NO.7

//7) Design java application using Collection in java such as Array List and Linked List

```
package practicles;

import java.util.ArrayList;
import java.util.Random;
import java.util.Scanner;

public class practicle_7 {
    public static void main(String[] args) {
        ArrayList<String> name; //declaring a String data type ArrayList
        Scanner sc = new Scanner(System.in);
        Scanner sc1 = new Scanner(System.in);
        int score = 0;
        int tmp_score = 0;
        System.out.println("enter the number of players. You cannot add less than
3 players and more than 10 players");

        int no_of_players;
        do{
            no_of_players = sc.nextInt();
            if(no_of_players<10 && no_of_players>3){
                System.out.println("you will add "+no_of_players+" of players");
            }
            else if(no_of_players>10){
                System.out.println("you can add only upto 10 players");
            }
            else if(no_of_players<3 ){
                System.out.println("you can not add less than 3 players");
            }
        }while (no_of_players>10 || no_of_players<3);

        name = new ArrayList<String>(); //initializing a ArrayList<String>
        System.out.println("enter the name of players : ");
        for(int i=0;i<no_of_players;i++){
            System.out.println("enter the "+i+"st name");
            name.add(i, sc1.nextLine());
        }

        Random r = new Random();
        int random_index;
        int n = 4;
        char toss;
        while (n>0){
            System.out.println("press n for tossing hands : ");
            toss = sc.next().charAt(0);
            if(toss == 'n'){
                random_index = r.nextInt(0,no_of_players);
                System.out.println("it is the turn of : "+name.get(random_index)
+ " at index number "+random_index);
                n=n-1;
            }
            else if(toss == 'x'){
```

```

        System.out.println("finished tossing");
        break;
    }
    else {
        System.out.println("it is invalid button");
    }
}
}
}
}

```

o/p:

enter the number of players. You cannot add less than 3 players and more than 10 players

3

enter the name of players :

enter the 0st name

Chaitanya

enter the 1st name

Sagar

enter the 2st name

Sushil

press n for tossing hands :

n

it is the turn of : Sagar at index number 1

press n for tossing hands :

n

it is the turn of : Chaitanya at index number 0

press n for tossing hands :

n

it is the turn of : Sushil at index number 2

press n for tossing hands :

n

it is the turn of : Sagar at index number 1

PRACTICEL NO. 8

//8) design gui based java application using AWT, Swing with Event Handling.
package practicles;

```
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.event.KeyEvent;
import java.awt.event.KeyListener;

//by using AWT components
public class practice_8 extends Frame implements KeyListener{
    Label l;
    TextField textField, textField2;
    Button button;

    public practice_8(){
        l = new Label("this is lable");

        textField = new TextField();
        textField.setBounds(300,400, 100, 20);

        textField2 = new TextField();
        textField2.setBounds(200, 100, 100, 20);

        button = new Button("ok");
        button.setBounds(200, 300, 30, 30);

        add(l);
        add(textField);
        add(textField2);
        add(button);

        setLayout(new FlowLayout());
        setSize(800, 800);
        setVisible(true);
    }
    public void buttonAction(){
        button.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                textField2.setText("pressed the button");
            }
        });
    }
    @Override
    public void keyTyped(KeyEvent e) {
        textField2.setText("you pressed button");
    }
    @Override
    public void keyPressed(KeyEvent e) {
        textField2.setText("you pressed button");
    }
}
```

```
@Override
public void keyReleased(KeyEvent e) {
    textField2.setText("you pressed button");
}
public static void main(String[] args) {
    practice_8 practice_8 = new practice_8();
    practice_8.buttonAction();
}
}
```

PRACTICEL NO. 9

//9) Design and implement a JDBC application

```
package practicles;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.Statement;

public class practicle_9 {
    public static void main(String[] args) {
        String name = "root";
        String pswd = "lxgiwyl0209";
        String url = "jdbc:mysql://localhost:3306/chaitanya";

        try{

            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, name, pswd);

            String q = "INSERT INTO persons (PersonId, LastName, FirstName,
Address, City)\n" +
                "VALUES (?, ?, ?, ?, ?)";

            PreparedStatement statement = con.prepareStatement(q);

            statement.setInt(1, 132);
            statement.setString(2, "Dharpale");
            statement.setString(3, "Chaitanya");
            statement.setString(4, "Bangalore");
            statement.setString(5, "Bangalore");
            statement.executeUpdate();

            System.out.println("successfully inserted");
        }
        catch (Exception e){
            System.out.println(e.getMessage());
        }
    }
}
```

PRACTICEL NO. 9

//10)Design and implement servlet application

```
import javax.servlet.http.*;
import javax.servlet.*;
import java.io.*;
public class DemoServlet extends HttpServlet{
    public void doGet(HttpServletRequest req,HttpServletResponse res)
        throws ServletException,IOException
    {
        res.setContentType("text/html");//setting the content type
        PrintWriter pw=res.getWriter();//get the stream to write the data

        //writing html in the stream
        pw.println("<html><body>");
        pw.println("Welcome to servlet");
        pw.println("</body></html>");

        pw.close();//closing the stream
    }
}
```

PRACTICLE 11

//11)Design and implement JSP application

//step 1 : create database table in oracle 10 database

1. **CREATE TABLE "USER432"**
2. **("NAME" VARCHAR2(4000),**
3. **"EMAIL" VARCHAR2(4000),**
4. **"PASS" VARCHAR2(4000)**
5. **)**
6. **/**

//step 2 : now create jsp page index.jsp

1. **<form action="process.jsp">**
2. **<input type="text" name="uname" value="Name..." onclick="this.value='"/>
**
3. **<input type="text" name="uemail" value="Email ID..." onclick="this.value='"/>
**
4. **<input type="password" name="upass" value="Password..." onclick="this.value='"/>
**
5. **<input type="submit" value="register"/>**
6. **</form>**

//step 3 : create process.jsp file

1. **<%@page import="bean.RegisterDao"%>**
2. **<jsp:useBean id="obj" class="bean.User"/>**
- 3.
4. **<jsp:setProperty property="*" name="obj"/>**
- 5.
6. **<%**
7. **int status=RegisterDao.register(obj);**
8. **if(status>0)**
9. **out.print("You are successfully registered");**
- 10.
11. **%>**

//step 4 : create a java file User.java

1. **package bean;**
- 2.

```
3. public class User {
4.     private String uname,upass,uemail;
5.
6.     public String getUname() {
7.         return uname;
8.     }
9.
10.    public void setUname(String uname) {
11.        this.uname = uname;
12.    }
13.
14.    public String getUpass() {
15.        return upass;
16.    }
17.
18.    public void setUpass(String upass) {
19.        this.upass = upass;
20.    }
21.
22.    public String getUemail() {
23.        return uemail;
24.    }
25.
26.    public void setUemail(String uemail) {
27.        this.uemail = uemail;
28.    }
29.
30. }
```

Step 5: provider.java class

```
1. package bean;
2.
3. public interface Provider {
4.     String DRIVER="oracle.jdbc.driver.OracleDriver";
```

```
5. String CONNECTION_URL="jdbc:oracle:thin:@localhost:1521:xe";
6. String USERNAME="system";
7. String PASSWORD="oracle";
8.
9. }
```

Step 6: connectionProvider.java class

```
1. package bean;
2. import java.sql.*;
3. import static bean.Provider.*;
4.
5. public class ConnectionProvider {
6.     private static Connection con=null;
7.     static{
8.         try{
9.             Class.forName(DRIVER);
10.            con=DriverManager.getConnection(CONNECTION_URL,USERNAME,PASSWORD);
11.        }catch(Exception e){}
12.    }
13.
14.    public static Connection getCon(){
15.        return con;
16.    }
17.
18. }
```

Step 7: register dao class

```
1. package bean;
2.
3. import java.sql.*;
4.
5. public class RegisterDao {
6.
7.     public static int register(User u){
```

```
8. int status=0;
9. try{
10. Connection con=ConnectionProvider.getCon();
11. PreparedStatement ps=con.prepareStatement("insert into user432 values(?,?,?)");
12. ps.setString(1,u.getUserName());
13. ps.setString(2,u.getUemail());
14. ps.setString(3,u.getUpass());
15.
16. status=ps.executeUpdate();
17. }catch(Exception e){}
18.
19. return status;
20. }
21.
22. }
```